

❏ Applying Security Patches via Bash

❏ 1. Automating Tasks Using Bash Scripts

Bash scripting is used to automate **routine system administration** and **user-level tasks**, improving consistency and reducing errors.

❏ Common Automatable Tasks

Task	Example
Backups	Automate rsync or tar operations
Log cleanup	Delete logs older than 30 days
Monitoring	Check disk space, services
File manipulation	Rename, move, compress
Cron jobs	Schedule periodic execution
User management	Add/delete users in bulk

📄 Sample Automation Script

bash

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```
#!/bin/bash # Backup /home every day src="/home" dest="/backup/home-$(date +%F).tar.gz" tar -czf "$dest" "$src" echo "Backup completed at $(date)"
```

📄 Automating with Cron

Edit crontab:

bash

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```
crontab -e
```

Example to run script daily at 1 AM:

```
bash
```

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```
0 1 * * * /scripts/backup.sh
```

📄 Log Rotation

Automated log management using:

- logrotate config
- Custom Bash scripts

☒ Cleanup Task Example

bash

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```
find /var/log -type f -name "*.log" -mtime +7 -exec rm -f {} \;
```

Deletes .log files older than 7 days.

❏ 2. Applying Security Patches

Keeping systems updated is a critical **security responsibility**.

❏ Why Automate Patching?

- Reduce vulnerability exposure
 - Ensure consistency across systems
 - Comply with policies
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❏ Patch Automation Tools

Tool	Description
apt-get	For Debian-based systems
yum or dnf	For RHEL/CentOS
unattended-upgrades	Auto installs security updates
Custom Bash	Cron or shell for updates

❏ Debian-based Script for Security Updates

bash

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```
#!/bin/bash # Apply security patches silently apt-get update -y apt-get upgrade -y apt-get autoremove -y
```

❏ RHEL/CentOS Patch Script

bash

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```
#!/bin/bash # Apply patches on CentOS yum update -y yum clean all
```

Or if using dnf:

bash

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```
dnf update -y
```

📄 Patching Logs

Always log patch activity:

```
bash
```

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```
yum update -y >> /var/log/yum-security-update.log 2>&1
```

📄 Best Practices for Patch Automation

- Test on staging servers
- Automate notification (email, logs)
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Use unattended-upgrades or **ansible automation** for large environments

- Exclude critical services if required

☒ Advanced Patch Management

- Combine with vulnerability scanners
- Integrate with SIEM
- Use **Ansible** or **Puppet** for multi-server automation